M. Rabiul Islam, PhD

Physiological Data Scientist | Signal Processing | Machine-Learning

Cleveland, Ohio • LinkedIn • Portfolio/Google Scholar

SUMMARY

Data Scientist | Expert in **Bio-Signal Processing**, **Epilepsy Research**, and **Machine Learning** | Innovator in Neuroinformatics and Precision Neurosurgery Solutions

EXPERIENCE

Postdoc Fellow August 2023 - Present

Cleveland Clinic, Ohio

Data Science Skills Used: Matlab, Python, R, Brainstorm, Free surfer, BrainNet Viewer, WinSPC, PuTTY, MNE Python, Decom Software, BrainNet Viewer

- **Collaborated with a multidisciplinary team**, utilizing Matlab, Python, R, and Brainstorm to drive advancements in neurological disease diagnosis and treatment, uncovering critical insights to inform clinical strategies.
- **Led AI-driven projects**, analyzing **Stereoelectroencephalography (SEEG)** data to extract key findings that enhanced diagnostic accuracy and improved patient outcomes.
- **Developed innovative data pipelines**, integrating signal processing and machine learning, resulting in improved early detection of neurological conditions and advancing clinical efficiency.
- Contributed to numerous scientific publications, presentations, and conferences sharing groundbreaking research findings and advancing knowledge in the field.

Postdoc Fellow December 2021 - November 2022

UTHealth, San Antonio, Texas

Data Science Skills Used: Python, R, R Shine, Spatial metabolomics

- Developed statistical models to predict disease biomarkers from biological data by leveraging techniques such as cluster analysis, linear regression analysis, generalized linear models, Cox proportional hazard models, multinomial logistic regression, random survival forest, and survival analysis. Visualized the data using R, RStudio, and R Shiny to create interactive applications.
- Developed data preprocessing and analysis pipelines that enhanced key performance indicators for patient outcome prediction by 15%.

Visiting Researcher April 2021-June 2021

Aizu University, Japan

Data Science Skills Used: Python, MATLAB

- Developed an artificial intelligent diagnostic tool for Parkinson's disease diagnosis, leveraging signal processing techniques and implementing feature extraction and classification algorithms.
- Utilized Python and MATLAB for preprocessing image data and visualizing the results.
- Leveraged signal processing and advanced algorithms to develop a Parkinson's disease diagnostic tool, enhancing accuracy and enabling precise treatment recommendations.

Research Assistant Professor April 2019 - March 2021

Tokyo University of Agriculture and Technology, Japan

Data Science Skills Used: Python, Matlab, Macintosh, Window, EEGLAB, EEG Brainstorm, LaTeX, keynote

- Developed AI techniques, including signal processing, feature engineering, and machine-learning methods, to analyze hospital data (ECOG, EEG, iEEG, SSVEP, ERP, MRI) for improved neurological insights.
- Constructed datasets and applied biomarker-based signal processing and machine-learning models to support clinical experts in making faster, data-driven decisions on neurological diseases.
- Published research in peer-reviewed journals and presented at international conferences, contributing to advancements in the field of neuroinformatics and precision medicine.

Assistant Professor January 2011 - March 2018

Department of Computer Science and Engineering, Pabna University of Science and Technology, Pabna, Bangladesh

- Involved in teaching and research at Undergraduate and Graduate Levels. Supervised 15 students' theses at Undergraduate, and Graduate Levels.
- Taught and completed practicum for various courses including Digital Signal and Image Processing, Digital Systems, Computer Vision, computer programming Python/R/C/C++, Database Management Systems, Microprocessors, Operating Systems, Image Processing and Computer Programming Lab etc.

EDUCATION

Doctor of Philosophy, Brain-computer Interface, Signal processing
Tokyo University of Agriculture and Technology, Tokyo, Japan October 2015-September 2018

- Development of a new machine learning algorithm application of Brain-Computer Interface (BCI). This area includes the development of Biosignal processing, feature-extraction, Machine-learning, and Statistical methods for robust BCI systems

Master of Science (MSc), Computer Science and Engineering
University of Rajshahi, Rajshahi, Bangladesh July 2010-September 2011

Bachelor of Science (BSc), Computer Science and Engineering
University of Rajshahi, Rajshahi, Bangladesh June 2004-June 2010

Honors & Awards

AES Fellowship 2024 **2024 to 2025**

- Awarded the prestigious AES Fellowship 2024 by the American Epilepsy Society to conduct advanced research in epilepsy and neuroinformatics. My research focuses on identifying seizure origins and networks through the analysis of stereo EEG (sEEG) data, aiming to develop quantitative signal biomarkers for precise seizure localization.

Monbukagakusho Fellowship from Japan Govt. **2014**

- Awarded the prestigious MEXT Fellowship from Japan Govt during my PhD studies to support advanced research in biosignal processing and EEG-based machine-learning methods for brain-computer interfaces. My research focused on developing innovative methodologies for analyzing neurological data. This fellowship recognized my contributions to various neurophysiological research areas, including EEG, SSVEP, and ECOG.

CERTIFICATIONS

- EEG/ERP Analysis with Python and MNE, Udemy
- Python for Data Science and Machine Learning python Bootcamp, Udemy
- Data Science and Machine Learning Bootcamp with R, Udemy
- Python Object Oriented Programming (OOPs), Udemy
- Deep Learning with Pytorch,Udemy
- Summary Statistics in Public Health, Johns Hopkins University Coursera
- Hypothesis Testing in Public Health, Johns Hopkins University Coursera

PUBLICATIONS

Journal paper:

1. M. Rabiul Islam et al., "Dynamic Stereoelectroencephalography-based Phase-Amplitude Coupling in Cingulate Epilepsy," *International Journal of Neural System*, 2025.
2. M. S. Yamamoto, K. Sadatnejad, T. Tanaka, **M. R. Islam**, F. Dehais, Y. Tanaka, F. Lotte,"Modeling complex EEG data distribution on the Riemannian manifold toward outlier detection and multimodal classification," *IEEE Transactions on Biomedical Engineering*, 2023.
3. **M. Rabiul Islam**, X. Zhao, Y. Miao, H. Sugano, and T. Tanaka, "Epileptic Seizure Focus Detection from Interictal Electroencephalogram: A Survey," *Cognitive Neurodynamics*, 2022.

4. K. M. Hassan, **M. R. Islam**, T. T. Nguyen, M. K. I. Moll,"Epileptic seizure detection in EEG using mutual information-based best individual feature selection," *Expert Systems with Applications*, Dec 2021. (**Corresponding author**)
5. S. Mahmood, J. Shin, I. Farhana, **M. R. Islam**, M. K. I. Molla,"Frequency Recognition of Short-Time SSVEP Signal using CORRCA based Spatio-Spectral Feature Fusion Framework," *IEEE access*, Dec 2021
6. M. K. I. Molla, S. Kumar, **M. Rabiul Islam**, S. Jungpil,"Trial Regeneration with Subband Signals for Motor Imagery Classification in BCI Paradigm," in *IEEE Access*, Dec, 2020.
7. M. S. Akter , **M. Rabiul Islam**, T. Tanaka, Y. Iimura, T. Mitsuhashi, H. Sugano, D. Wang, M. K. I. Molla,"Statistical features in high-frequency bands of interictal iEEG work efficiently in identifying the seizure onset zone in patients with focal epilepsy," *Entropy*, 20(12), 2020.
8. M. K. I. Molla, Kazi Mahmudul Hasan, **M. Rabiul Islam** and Toshihisa Tanaka,"Graph Eigen Decomposition Based Feature Selection Method for Epileptic Seizure Detection Using Electroencephalography," *Sensors*, 2020, 20, 4639.
9. M. K. I. Molla, A. A. Shiam, **M. Rabiul Islam**, T. Tanaka,"Discriminative Feature Selection Based Motor Imagery Classification Using EEG Signal," *IEEE Access*, 2020.
10. M. S. Akter, **M. Rabiul Islam**, Y. Iimura, H. Sugano, K. Fukumori, D. Wang, T. Tanaka, A. Cichocki, "Multiband entropy-based feature-extraction method for automatic identification of epileptic focus based on high-frequency components in interictal iEEG," *Scientific Reports* (Nature), Vol. 10, pp. 7044, April, 2020.
11. M. K. I. Molla, N. Morikawa, **M. Rabiul Islam** and Toshihisa Tanaka,"Data-adaptive Spatiotemporal ERP Cleaning for Single-trial BCI Implementation,"*IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 26, pp. 1334-1344, May 2018.
12. **M. Rabiul Islam**, Toshihisa Tanaka and M. K. I. Molla,"Multiband Tangent Space Mapping and Feature Selection for Classification of EEG during Motor Imagery,"*Journal of Neural Engineering*, Vol. 15, no. 4, pp. 046021, May 2018.
13. N. Morikawa, Toshihisa Tanaka and **M. Rabiul Islam** ,"Complex Sparse Spatial Filter for Decoding Mixed Frequency and Phase Coded Steady-State Visually Evoked Potentials," *Journal of Neuroscience Methods*, Vol. 304, pp. 1-10, April 2018.
14. **M. Rabiul Islam**, Toshihisa Tanaka, Masaki Nakanishi and M. K. I. Molla,"Unsupervised Frequency-Recognition Method of SSVEPs using a Filter Bank Implementation of Binary Subband CCA,"*Journal of Neural Engineering*, Vol. 14, no.2, pp. 026007, Jan 2017.
15. M. S. Islam, **M. Rabiul Islam**, A. Hossain, A. Ferworn and M. K. I. Molla,"Subband Entropy Based Features for Clothing Invariant Human Gait Recognition,"*Advanced Robotics* (Taylor & Francis), vol. 31, pp. 519-530, Dec 2016.
16. M. R. Al-Mahfuz, **M. Rabiul Islam**, Keikichi Hirose, M. K. I. Molla,"Artifact Suppression and Analysis of Brain Activities with EEG Signals,"*Neural Regeneration Research*, Vol. 16, pp. 1500-1513, June 2013.

17. M. S. Akter, **M. Rabiul Islam**, B. K. Pramanik and M. S. Islam,"Separation of Ocular Artifacts from EEG Signal using Noise Assisted Bi-variate Adaptive Filtering,"*Journal of Bio-Science and Bio-Technology*, vol. 5, pp. 117-128, Aug 2013.
18. M. K. I. Molla, **M. Rabiul Islam**, Toshihisa Tanaka and Tomasz Rutkowski,"Artifact suppression from EEG signals using data adaptive time domain filtering,"*Neurocomputing*, Vol.97, pp.297-308, Nov. 2012.
19. **M. Rabiul Islam**, S. Das, Keikichi Hirose and M. K. I. Molla,"Analysis of ECG signals using data adaptive time domain filtering for cardiovascular disease diagnosis,"*Advances in Adaptive Data Analysis* (World Scientific), Vol. 04, 30 August 2012.
20. **M. Rabiul Islam**, M. R. Al-Mahfuz and M. K. I. Molla,"Multiband Prediction Model for Financial Time Series with Multivariate Empirical Mode Decomposition,"*Discrete Dynamics in Nature and Society*, Vol. 2012, November 2011.

Conference papers

1. M. S. Yamamoto, K. Sadatnejad, T. Tanaka, **M. Rabiul Islam**, Y. Tanaka and F. Lotte, "Detecting EEG outliers for BCI on the Riemannian manifold using spectral clustering,"in *42nd Annual International Conferences of the IEEE Engineering in Medicine and Biology Society* (EMBC 2020), Montreal, Canada, 2020, pp. 438-441.
2. M. S. Akter, **M. Rabiul Islam**, K. Fukumori, Yasushi Iimura, Hidenori Sugano and Toshihisa Tanaka,"Automatic Detection of Epileptic Focus in Ripple and Fast Ripple Bands of Interictal iEEG based on multi-band analysis,"in *International Conference on Artificial Intelligence in Information and Communication* (ICAIIC 2020), Fukuoka, Japan, 2020, pp. 490-493.
3. K. M. Hassan. M, M. K. I. M, **M. Rabiul Islam**, and Toshihisa Tanaka,"Epileptic Seizure Detection from EEG Signals Using Multiband Features with Feedforward Neural Network,"in *International Conference on Cyberworlds* (CW 2019), Kyoto, Japan, 2019, pp. 231-238
4. Abdullah Al Shiam, Md. Khademul Islam Molla, **Md. Rabiul Islam** and Toshihisa Tanaka, "Electroencephalography Based Motor Imagery Classification Using Unsupervised Feature Selection,"in *International Conference on Cyberworlds* (CW 2019), Kyoto, Japan, 2019, pp. 239-246.
5. M. S. Akter, **M. Rabiul Islam**, Toshihisa Tanaka, Kosuke Fukumori, Yasushi Iimura, and Hidenori Sugano,"Automatic identification of epileptic focus on high-frequency components in interictal iEEG,"*8th International Congress on Advanced Applied Informatics* (IIAI-AAI 2019), Toyama, Japan, 2019, pp. 1075-1076
6. A. S. Musa, **M. Rabiul Islam**, M. K. I. Molla,"Motor Imagery Classification using Subband Tangent Space Mapping,"Proc. of *the Int. Conf. on Computer and Information Technology* (ICCIT 2017), Dhaka, 2017, pp. 1-5.
7. **M. Rabiul Islam**, Toshihisa Tanaka, M. K. I. Molla, and M. S. Akter,"Classification of Motor Imagery BCI Using Multiband Tangent Space Mapping,"Proceedings of the *22nd IEEE Int. Conf. on Digital Signal Processing* (DSP 2017), London, 2017, pp. 1-5.

8. M. K. I. Molla, M. A. Khan and **M. Rabiul Islam**,"Artifact Suppression from Electroencephalography Signals Using Stationary Subspace Analysis,"in *19th International Conference on Computer and Information Technology (ICCIT 2016), Dhaka, 2016, pp. 252-256*
9. **M. Rabiul Islam**, Toshihisa Tanaka, Masaki Nakanishi and M. K. I. Molla,"Frequency Recognition of Steady-state Visually Evoked Potentials Using Binary Subband Canonical Correlation Analysis With Reduced Dimension of Reference Signals,"in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2016)*, Shanghai, China, 2016, pp. 769-773.
10. **M. Rabiul Islam**, Toshihisa Tanaka, M. K. I. Molla and M. S. Akter,"Frequency Recognition of SSVEP-Based BCI Using Reference Signal With Dominant Frequency,"in *Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA 2015)*, Hong Kong, China, 2015, pp. 971-974.
11. **M. Rabiul Islam**, Toshihisa Tanaka, N. Morikawa and M. K. I. Molla,"Frequency Recognition for SSVEP–based BCI with Data Adaptive Reference signals,"Proceedings of *IEEE Digital Signal Processing* (DSP), Singapore, 2015.
12. M. K. I. Molla, Toshihisa Tanaka, K. Osa and **M. Rabiul Islam**,"EEG Signal Enhancement Using Multivariate Wavelet Transformation: Application to Single-Trial Classification of Event-Related Potentials,"in *IEEE International Conference on Digital Signal Processing (DSP 2015)*, Singapore, 2015, pp. 799-803.
13. M. S. Akter, **M. Rabiul Islam**, Md. Khademul Islam Molla and R. Mahfuz,"Speech Enhancement using EMD-Based Post Filtering with Spectral Weighting,"in *2nd International Conference on Advances in Electrical Engineering (ICAEE 2013)*, Dkaka, Bangladesh, 2013, pp. 29-32.
14. M. S. Islam, **M. Rabiul Islam**, M. S. Akter, M. A. Hossain and Md. Khademul Islam Molla,"Window Based Clothing Invariant Gait Recognition,"in *2nd International Conference on Advances in Electrical Engineering (ICAEE 2013)*, Dkaka, Bangladesh, 2013, pp. 411-414.
15. **M. Rabiul Islam**, S. Ahmad, Keikichi Hirose and M. K. I. Molla, "Data adaptive analysis of ECG signals for cardiovascular disease diagnosis,"*in Proceedings of IEEE International Symposium on Circuits and Systems (ISCAS 2010),* Paris, France, 2010, pp. 2243-2246.

Abstracts

1. **M. Rabiul Islam** et al. "Epileptiform State Dependence of Band-Specific Phase–Amplitude Coupling in sEEG Data from Refractory Temporal Lobe Epilepsy Patients," accepted for presentation at the American Society for Stereotactic and Functional Neurosurgery Biennial Meeting, 2026.

2. **M. Rabiul Islam** et al. "Dynamic Band-Specific Frequency Coupling for Patients with Cingulate Epilepsy: A Stereo-EEG Study," American Epilepsy Society Annual Meeting, 2025.

3. **M. Rabiul Islam** et al. “Phase-amplitude Coupling in Cingulate Epilepsy Recordings: An Seeg-based Approach,” American Epilepsy Society Annual Meeting, 2024.

4. M. S. Yamamoto, K. Sadatnejat, **M. Rabiul Islam**, F. Lotte, Toshihisa Tanaka,"Reliable outlier detection by spectral clustering on Riemannian manifolds of EEG covariance matrix," 8th International BCI meeting, 2020, Brussels, Belgium.